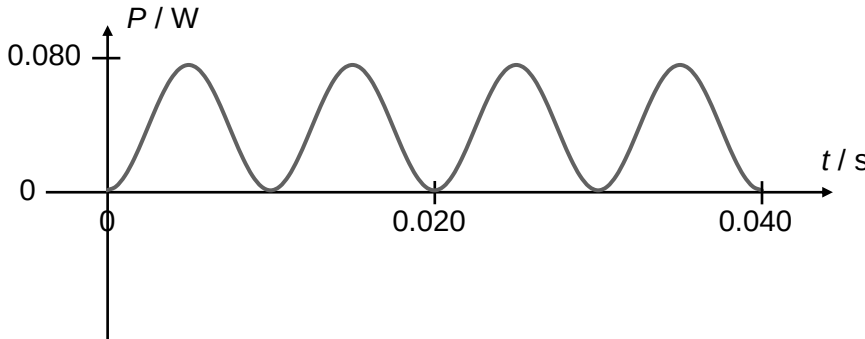


6	(a)	$V_{r.m.s.} = \sqrt{\frac{2^2(0.002) + 1^2(0.002)}{0.010}} = 1.0 \text{ V}$ Steady voltage of 1.0 V will produce the same heating effect as $V_{r.m.s.}$ of 1.0 V.	M1 A1
		<i>Marker's comments: Common mistakes like without taking the square root value of the mean square voltage was observed.</i>	

	(b)	Transmission of electrical energy at high voltage means that the current is low according to $P = IV$. Power loss through joule heating (I^2R) is hence lowered as less electrical energy is dissipated as heat in the cables of resistance R.	B1 B1
		<i>Marker's comments: no mention of power loss as heat dissipation was very common among the answers given by candidates.</i>	
	(c)(i)	$\frac{V_s}{V_p} = \frac{N_s}{N_p}$ $V_s = 71 \times 6.5 \times 10^{-3} = 0.46 \text{ V}$	A1
	(c)(ii)	 Correct shape: $0.080 \sin^2[(2\pi f)t]$. Correct labelling of values: peak power and period.	B1 B1
		<i>Marker's comments: Most answers given was a sine curve instead of sine square curve.</i>	
	(c)(iii)	1. In the forward biased direction, the diode has no resistance. Current flows downwards through resistor R. In the reverse biased direction, diode has infinite resistance. There is no current flowing through resistor R.	B1 B1
	(c)(iii)	2. In the <u>forward biased direction</u>, there is a <u>half-wave sinusoidal voltage</u> across resistor R, having the same frequency as that of the input voltage. In the reverse biased direction, there is no voltage across resistor R.	B1
		<i>Marker's comments: No mention of why there was a current in one direction and no current in the other direction.</i>	